Exploring LLMs for EEG Classification through Active Prompting with Purity-Guided Unique-Round Example Selection (PURE)

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***Abstract*—Large language models (LLMs) have demonstrated impressive in-context learning (ICL) capabilities, particularly in few-shot prompting scenarios. However, most studies have focused on natural language processing (NLP) and computer vision (CV), with limited exploration of tasks based on complex physiologic signals such as electroencephalogram (EEG) signal classification. To bridge this gap, we propose a purity-guided active prompting approach, which integrates domain-specific EEG feature extraction with an active learning-based demonstration selection strategy. PURE identifies a compact set of prototypical examples that effectively activate the few-shot capabilities of LLMs without requiring gradient-based parameter updates, and is compatible with both open-source and closed-source models. Unlike the conventional framework that involves random selection of demonstration examples on a per-instance basis for each test sample, PURE actively identifies a small set of highly representative EEG samples just once from the training data, significantly reducing computational overhead. Extensive experiments on seven EEG datasets and three paradigms demonstrate that PURE consistently outperforms random selection baseline, achieving superior accuracy and robustness. Furthermore, performance improves with more powerful LLMs (e.g., DeepSeek-V3), and in some cases, even surpasses state-of-the-art supervised models.**

***Index Terms*—EEG, BCI, LLMs, active learning, prompt engineering, in-context learning, few-shot prompting, knowledge fusion, motor imagery, sleep-stage classification, seizure detection**

I. Introduction

Transformer-based large language models (LLMs) [1] have achieved remarkable progress across a wide range of natural language processing (NLP) tasks. As model size increases, LLMs exhibit strong in-context learning (ICL) capabilities, enabling them to perform downstream tasks by conditioning on a few demonstration examples at inference time, without any parameter updates [2].

A brain-computer interface (BCI) [3], [4] establish a direct communication pathway between the human brain and external devices, allowing users to interact with their surroundings using brain signals that reflect cognitive states or intentions [5]. Among various BCI modalities, electroencephalography (EEG), which records electrical activity from the scalp, is the most widely used input signal due to its low cost, non-invasiveness, and convenience.

Although ICL has been extensively studied in NLP and computer vision (CV), its application in BCI tasks, especially those based on EEG signals, remains largely unexplored. To bridge this gap, it is essential to develop effective strategies for embedding complex numerical data into textual prompts that can be processed by LLMs.

Prior efforts have investigated such integration in domains such as tabular data [6] and medical records [7], where numerical values often possess clear semantic meaning and can be easily verbalized. In contrast, EEG signals lack inherent semantic interpretability. For example, the value “16” in a clinical record may represent a patient’s age, while in EEG data it may simply denote a signal amplitude at a specific time point, devoid of symbolic meaning.

Moreover, EEG signals are high-dimensional and temporally dense. A two-second EEG trial sampled at 256 Hz across 8 channels yields 4,096 raw values. Directly inputting such dense and unstructured data into an LLM results in excessive token consumption and limits the model’s capacity to extract meaningful patterns. Additionally, EEG signals are non-stationary, exhibit substantial inter-subject variability, and suffer from low signal-to-noise ratios. These challenges have left the direct application of LLMs to EEG largely unexplored.

To address these challenges, we propose a simple yet effective framework that incorporating domain knowledge to extract discriminative and compact features, so that they can be effectively processed by LLMs, and selects highest intra-class consistent examples by the proportion of same-label neighbors within a k-nearest neighborhood. Experimental results on seven EEG datasets across three paradigms demonstrate that our method consistently achieves better and more stable performance than random selection.

In summary, our contributions in this paper are:

1) To the best of our knowledge, this work is the first to directly explore the feasibility of using LLMs for classifying EEG signals via in-context few-shot prompting, thereby extending the application of LLMs to complex, low-semantic numerical domains beyond natural language.
2) We incorporate domain-specific knowledge to extract features from EEG signals. This not only reduces the number of input tokens substantially but also improves classification performance, making the LLM-

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based prompting pipeline more effective and computationally feasible.

3) Unlike prior work in NLP that performs example selection per test instance, we introduce a fixed example set that is selected once from the training pool and reused across all test samples. This strategy significantly reduces computational cost and inference latency.
4) We propose a purity-guided active prompting method named PURE, which is validated on seven EEG datasets spanning three paradigms. Experimental results demonstrate that PURE consistently outperforms random selection in terms of performance and stability.

The remainder of this paper is organized as follows: Section II introduces related work. Section III proposes PURE. Section V presents the experimental results. Finally, Section VI draws conclusions.

II. Related Works

A. EEG-based Classification

EEG-based classification encompasses a wide range of tasks, including motor imagery (MI) [8], sleep stage classification [9], and epilepsy detection [10]. Traditional approaches typically rely on feature extraction methods such as common spatial pattern (CSP) [11], frequency-domain features (e.g., power spectral density), time-domain features (e.g., amplitude statistics, zero-crossing rate), and nonlinear features (e.g., entropy, complexity measures) [12], followed by classifiers such as logistic regression (LR) [13] or support vector machines (SVM) [14].

To enhance EEG feature extraction, Ang *et al.* [15] proposed the filter bank CSP (FBCSP) method, which applies CSP to multiple frequency bands and automatically selects the most discriminative features for classification. Lotte *et al.* [16] introduced regularized CSP to improve the robustness of feature extraction.

More recently, deep learning models have achieved remarkable success in EEG decoding by integrating feature extraction and classification into a single end-to-end architecture [17], [18]. In particular, convolutional neural networks (CNNs) have demonstrated superior performance in MI classification.

For example, Schirrmeister *et al.* [19] proposed ShallowCNN, which uses temporal and spatial convolutional filters to extract relevant features for MI decoding. Lawhern *et al.* [20] introduced EEGNet, a compact model employing depthwise and separable convolutions to significantly reduce the number of parameters while achieving strong performance across various BCI tasks, including MI classification. Jiang *et al.* [21] proposed CSP-Nets, a hybrid model that integrates knowledge-driven CSP filters with data-driven CNNs to enhance MI classification performance.

For sleep stage classification, Supratak *et al.* [22] proposed DeepSleepNet, which combines CNN and bidirectional LSTM architectures and achieved state-of-the-art (SOTA) performance on the Sleep-EDF dataset. For seizure prediction, Golmohammadi *et al.* [23] utilized LSTM to effectively capture the temporal dynamics of EEG signals, thereby improving prediction accuracy.

A few recent studies have also explored the use of large-scale models for EEG-based classification. Jiang *et al.* [24] introduced a 0.369B-parameter model trained on a large-scale EEG corpus spanning multiple paradigms and tasks, representing one of the most extensive models in the EEG domain. However, due to the limited size of publicly available EEG datasets, model scalability remains constrained, and its performance and usability in downstream applications still fall short compared to LLMs. Due to the modality gap, no existing study has directly applied LLMs to EEG classification tasks via prompt engineering.

B. Active Learning

Active learning (AL), a subfield of machine learning, aims to reduce annotation costs by enabling models to proactively select the most informative unlabeled instances for labeling. This strategy enhances model performance while minimizing labeling effort.

Initial research in AL primarily emphasized uncertainty-based sampling methods, which prioritize samples near the decision boundary or those associated with high classification uncertainty [25]. For example, Lewis *et al.* [26] demonstrated that uncertainty sampling significantly reduces annotation costs in text classification tasks, outperforming both random and relevance sampling methods, particularly when training data is limited.

Beyond uncertainty, representativeness has emerged as a complementary criterion. Strategies such as clustering-based sampling [27] and density-weighted uncertainty sampling [28] seek to select samples that are both uncertain and representative, avoiding outliers.

Multi-criteria approaches further enhance AL performance. Shen *et al.* [29] applied informativeness, representativeness, and diversity jointly to named entity recognition. Huang *et al.* [30] introduced QUIRE, which selects samples by minimizing maximum model change (MMC), effectively balancing informativeness and representativeness. Cai *et al.* [31] proposed a MMC strategy tailored to SVM, although it is sensitive to initial parameters. He *et al.* [32] formulated a batch-mode AL method combining uncertainty, representativeness, and diversity. Their method quantifies informativeness as the product of uncertainty and representativeness, clusters samples via kernel k-means, and selects cluster centers for labeling.

In regression tasks, Burbidge *et al.* [33] applied a query-by-committee (QBC) strategy based on model disagreement, though inaccurate model assumptions may degrade performance. Wu *et al.* [34] developed the RD method, a sequential pool-based AL approach that incorporates representativeness and diversity during both initialization and iteration. RD can be combined with existing regression-based AL methods such as QBC, MCM, and Greedy Sampling (GS) to further enhance performance. Wu *et al.* [35] also advocated GS for regression, focusing on maximizing diversity.

Despite its widespread application in selecting training samples for traditional machine learning models, AL has been rarely explored for prompt selection in LLMs. Whether the selection criteria developed for conventional models remain effective in few-shot prompting scenarios with LLMs remains an open question.

C. Demonstration Selection for In-Context Learning

With the scaling of both model and data sizes [2], [36]–[38], LLMs have demonstrated the ability to perform ICL, a paradigm wherein models learn tasks from a few example demonstrations without parameter updates [39].

Despite its simplicity, the effectiveness of ICL is highly sensitive to the selection and arrangement of demonstration examples. Inferior choices in content or ordering can degrade performance and introduce unintended biases.

Most existing approaches to demonstration selection adopt a per-instance strategy, where demonstrations are selected independently for each test instance. This leads to two key selection objectives: similarity and diversity [40].

Liu *et al.* [41] proposed selecting demonstrations based on similarity by retrieving the k nearest neighbors of the test instance using sentence encoders in the embedding space. However, this approach does not consider diversity, which may result in homogeneous and redundant examples. Moreover, in classification tasks, kNN-based selection often biases the model toward predicting the majority label among the selected examples, effectively reducing the LLM to a kNN classifier and diminishing its unique generalization capabilities.

To address the issue of homogeneity, Li *et al.* [42] introduced clustering-based retrieval, where the sample pool is first clustered, and the nearest instance to the query is selected from each cluster. While this approach increases diversity, it may include examples that lack sufficient similarity to the query.

To better balance similarity and diversity, iterative retrieval methods have been proposed. These methods begin by selecting the example most similar to the query and then sequentially add examples by considering both their similarity to the query and their diversity relative to previously selected samples. Scarlatos *et al.* [43] trained an LSTM-based retriever within a reinforcement learning framework. During inference, the retriever selects an initial demonstration by processing the query, then updates the query representation using the LSTM's hidden state to incorporate prior demonstrations. This iterative process continues until k demonstrations are selected [40].

In computer vision (CV), this idea has been extended by assigning similarity-based scores (e.g., cosine distance) to candidate examples to quantify their relevance for inclusion in the prompt [44].

In contrast to per-instance approaches, Hong *et al.* [45] proposed a two-step method: first, a diverse and representative subset is selected for annotation; then, task-specific demonstrations are retrieved from this labeled pool at inference.

Overall, existing work on ICL demonstration selection has primarily focused on NLP and CV tasks, with strategies largely centered on balancing similarity and diversity among examples. While these criteria have shown effectiveness in certain contexts, their applicability to other domains and modalities remains underexplored.

III. Method

This section introduces our proposed method, PURE. Specifically, we start with a large pool of labeled training samples and select a compact subset to serve as in-context examples within the prompt, aiming to fully leverage the few-shot capabilities of the LLM and maximize its performance on the test set. The overall procedure is illustrated in Figure 1.

A. Domain Knowledge Driven Feature Extraction

Given the complexity of raw EEG signals, it remains necessary to extract compact and discriminative handcrafted features after standard preprocessing. This step is crucial not only for enhancing task-relevant signal characteristics but also for reducing the input dimensionality to meet the token-length constraints of LLMs. The resulting feature vectors are then used as in-context demonstrations in the LLM prompt.

To this end, domain knowledge is incorporated, and feature extraction methods are selected based on the specific paradigm. For MI classification, which primarily elicits event-related desynchronization (ERD) and synchronization (ERS) patterns in the μ (8–13 Hz) and β (13–30 Hz) frequency bands over the sensorimotor cortex, which particularly around electrodes C3 and C4, we adopt a feature extraction pipeline inspired by the FBCSP framework [15]. Specifically, we apply a band-pass filter over the 8–24 Hz range, divide the spectrum into four 4-Hz sub-bands, and extract two CSP features from each sub-band. This approach significantly improves classification performance.

For sleep stage classification, a wide range of EEG feature extraction techniques are available, spanning the time, frequency, and time-frequency domains. In the time domain, standard statistical features such as amplitude measures and zero-crossing rates are employed. In the frequency domain, power spectral density and band power features are commonly used. Time-frequency domain approaches include wavelet transform, short-time Fourier transform, empirical mode decomposition, and Wigner–Ville distribution. Since our primary goal is to evaluate the impact of demonstration selection strategies (random vs. active), we focus on delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), spindle (12 - 14 Hz) and beta (14–30 Hz) bands, which have been shown to correspond to sleep stages W, N1–N3, and REM in prior EEG literature, and adopt a simple yet effective feature representation using multi-band power spectral density features.

For epileptic seizure detection, a wide variety of EEG feature extraction techniques have been investigated across the time, frequency, and time-frequency domains. Through extensive experimentation, Boonyakitanont *et al.* [46] demonstrated that features such as variance, energy, nonlinear energy, and Shannon entropy computed directly from raw EEG signals, as well as variance, energy, kurtosis, and line length derived from wavelet coefficients, are effective in capturing seizure-related patterns. Among these, amplitude-based features were found to be particularly discriminative. Motivated by these findings, we adopt a practical representation by extracting multi-band signal power features directly from the raw EEG signals across standard frequency bands.

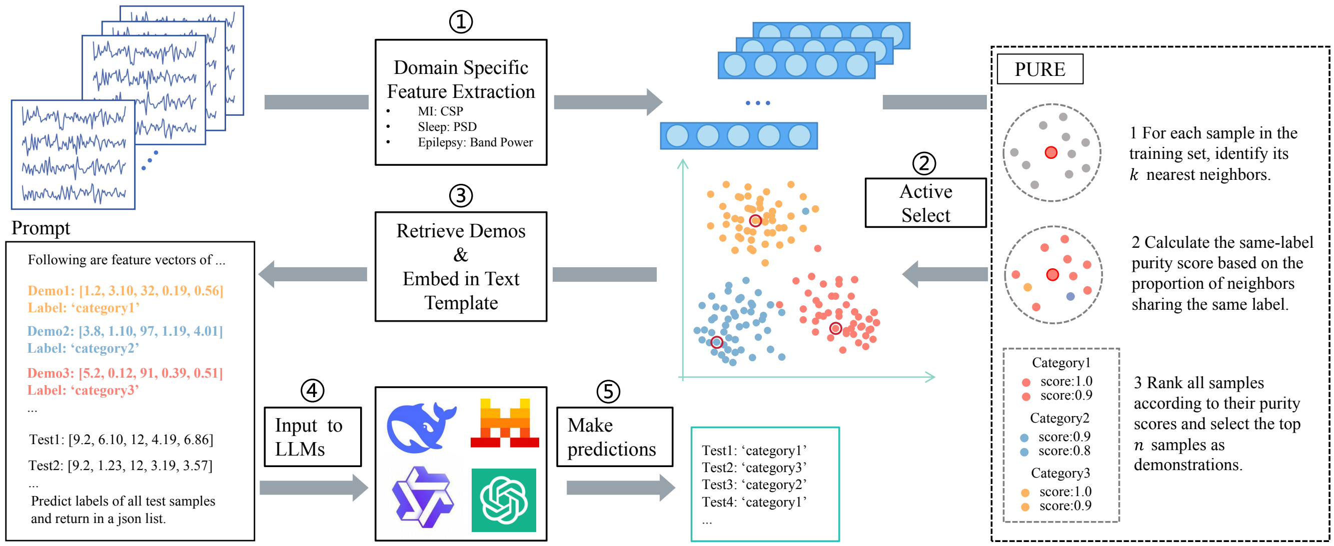

Fig. 1. Flowchart of EEG-based classification using LLM few-shot prompting.

B. Purity-Based Unique-Round Example Selection (PURE)

After obtaining compact feature vectors, the next step is to select a small subset of examples that can most effectively activate the few-shot learning capabilities of the LLM. However, as previously discussed, it remains unclear whether principles from AL are directly applicable to EEG-based classification with LLMs. To investigate this, we conducted a series of experiments to evaluate the effectiveness of different example selection strategies.

Surprisingly, and contrary to conventional assumptions, we found that the most informative samples, such as those located near the decision boundary and associated with high prediction uncertainty, did not yield the best performance. Instead, the most confidently predicted samples, which are typically less informative in traditional active learning, consistently led to higher classification accuracy. This observation aligns with our hypothesis that high-uncertainty examples may introduce ambiguity and provide less effective guidance during inference. In contrast, prototypical and easily classifiable examples better represent the underlying characteristics of each class, thereby supporting more accurate model predictions.

Based on this insight, we propose PURE (Purity-Based Unique-Round Example Selection), a simple yet effective method for selecting in-context examples in EEG-based tasks. For each training sample, we compute its k nearest neighbors and calculate the proportion of neighbors sharing the same class label—a metric we refer to as local same-label purity. As illustrated in Algorithm 1, all training samples are ranked based on their purity scores, and the top-n most "pure" samples are selected as in-context examples. This strategy not only favors the most representative samples within each class but also inherently avoids outliers, as samples surrounded by mixed-class neighbors receive lower purity scores and are less likely to be chosen.

Although PURE does not explicitly enforce diversity, we empirically observe that augmenting it with greedy diversity sampling leads to performance degradation. This suggests that purity alone is a more suitable criterion in this context.

Importantly, PURE adopts a one-time selection strategy, in which a fixed set of in-context examples is selected once from the training set and reused across all test instances. Compared to per-instance selection, this approach offers significant advantages in terms of computational efficiency and scalability:

1) *Reduced computational overhead*. It avoids repeated example selection for each test input, eliminating the need for instance-wise retrieval.
2) *Improved batch efficiency*. Since all test samples share the same in-context examples, multiple test inputs can be batched into a single prompt, enabling parallel predictions with a single API call. In contrast, per-instance selection requires constructing a unique prompt for each test input, yielding only one prediction per query.

C. Prompt Design

Finally, the selected examples and their corresponding labels are combined with the task description, background context, and explicit output instructions to construct a complete in-context prompt, which is then provided to the LLM alongside the test input.

In our prompt design, we rigorously follow best practices in prompt engineering to ensure clear and effective communication with the LLM [47]. Specifically, the task description is kept concise, unambiguous, and placed at the beginning of the prompt to set the model's objective. The context section includes relevant background information and a few representative input–output pairs, enabling the model to infer the desired mapping pattern from demonstrations.

Explicit output instructions are also incorporated to guide the model toward producing structured and predictable responses, which are critical for downstream evaluation and integration. These instructions specify both content constraints (e.g., label formats, output options) and style constraints (e.g., use of quotation marks, line breaks). In addition, we apply prompt formatting techniques such as section delimiters (e.g., "###" or newline markers), consistent indentation, and token normalization to delineate boundaries between different components—such as descriptions, examples, and the query—thereby improving input clarity and model interpretability. A typical prompt structure used in our study is showed in Fig 2.

To prevent prompt overload or truncation, all components are optimized for brevity without compromising informativeness. This balance is particularly important when working with token-limited APIs or long test inputs. Our final prompt format is compatible with both open-source and API-based LLMs and is designed to be reusable across batches of test inputs for efficient inference.

Algorithm 1: Purity-based Unique-Round Example Selection (PURE)

Input: N labeled samples, $\{\mathbf{x}_n\}_{n=1}^{N}$, where $\mathbf{x}_n \in \mathbf{R}^d$;
C, number of classes;
M, number of samples to select;
K, neighborhood size

Output: A list of selected sample indices with balanced class distribution L.

Ensure M is divisible by C;
foreach *sample* $n = 1$ *to* N **do**
  Compute distances between $\mathbf{x}_n$ and all other samples;
  Find K nearest neighbors of $\mathbf{x}_n$ (excluding itself);
  Compute same-label purity score:

$$\text{purity}(\mathbf{x}_n) = \frac{\text{Number of same-label neighbors}}{K}$$

end
Group all samples by class label;
Sort each group in descending order of purity score;
foreach *class* $c = 1$ *to* C **do**
  Select top M/C samples with highest purity;
  Add their indices to output list L;
end
return *Selected sample indices list L.*

```
### Task Description ###
Given a set of feature vectors of EEG samples,
determine whether the category of the sample to be
predicted is A or B.

### Demonstrations ###
Features: [0.23, 1.45, 0.87, ..., 0.91]
Label: A
Features: [1.14, 0.76, 1.09, ..., 0.67]
Label: B
...

### Samples to be predicted ###
Features: [0.41, 1.30, 0.99, ..., 0.80]
Features: [1.08, 0.64, 1.05, ..., 0.72]
Features: [0.27, 1.38, 0.90, ..., 0.76]
Features: [0.37, 4.38, 2.90, ..., 0.16]
...

### Instructions ###
Predict the label of the following input and return in a
JSON list.
```

Fig. 2. A typical prompt structure used in our experiments.

IV. Datasets

This section introduces five MI datasets, one sleep dataset, and one epilepsy dataset used in our experiments.

A. MI Datasets

All MI datasets used in our experiments are binary classification tasks and were obtained using the MOABB framework [48]. Specifically, two datasets from BCI Competition IV (BNCI2014001 and BNCI2014004) [49] contain left-vs-right hand imagery, while the other three: BNCI2014002, BNCI2015001, and Weibo2014, contain right-hand vs. feet imagery.

Their experimental paradigms were similar: In each session, a subject sat comfortably in an armchair positioned in front of a computer screen. At the beginning of each trial, a fixation cross appeared on a black background, accompanied by a short auditory warning tone. After a brief delay, a directional cue in the form of an arrow pointing left, right, down, or up appeared on the screen, corresponding to one of the four MI classes: left hand, right hand, feet, or tongue. This cue remained visible throughout the task period and served as a prompt for the subject to begin the corresponding MI task. Subject performed the imagery task until the fixation cross disappeared. Each trial ended with a brief black screen before the next trial began [49].

Notably, for BNCI2015001, the last 3 subjects were excluded due to poor performance [50]. And for Weibo2014, only right-hand and feet trials were used for binary classification in this study. From the original 60 usable channels, we selected 21 MI-related channels: FC3, FC1, FCz, FC2, FC4, C5, C3, C1, Cz, C2, C4, C6, CP3, CP1, CPz, CP2, CP4, P3, P1, P2, and P4.

B. Sleep Datasets

The Sleep-EDF Expanded (Sleep-EDFx) dataset is intended for sleep research, particularly for sleep stage classification. It contains 197 full-night polysomnographic (PSG) recordings, including EEG (Fpz–Cz and Pz–Oz), EOG, EMG, and event markers. Corresponding hypnograms (sleep stage annotations) were manually scored by trained technicians following the Rechtschaffen and Kales manual and are provided alongside the recordings.

In our experiments, we use only EEG channels for a five-stage sleep classification task. Due to the large volume of data,

TABLE I
SUMMARY OF THE SEVEN EEG DATASETS.

BCI Paradigm	Dataset	Number of Subjects	Number of Channels	Sampling Rate (Hz)	Trial Length (seconds)	Number of Trials per Subject	Class Labels
MI	BNCI2014001	9	22	250	4	288	left hand, right hand
	BNCI2014004	9	3	250	4.5	680-760	left hand, right hand
	BNCI2014002	14	15	512	5	160	right hand, both feet
	BNCI2015001	12(9)	13	512	5	400/600	right hand, both feet
	Weibo2014	10	60(22)	200	4	140, 160	right hand, both feet
Sleep	Sleep-EdfX	9	2	100	30	1025	W, N1, N2, N3, REM
Epilepsy	CHB-MIT	10	23	256	3	76-618	seizures, non-seizures

we selected recordings from first seven subjects, using one night of data per subject.

During the signal processing and feature extraction stages, EEG data from the Sleep-EDFx dataset were first bandpass filtered between 0.5 Hz and 35 Hz. Since the recordings span an entire night and include prolonged wake periods at the beginning and end, only the first and last 30 minutes of wakefulness were retained to reduce redundancy. Sleep stages were labeled according to the AASM standard and grouped into five classes: Wake (W), N1, N2, N3, and REM. Each trial was segmented into non-overlapping 30-second epochs.

To characterize the spectral properties of the EEG signals, we extracted frequency-domain features using Welch's method for power spectral density (PSD) estimation. For each epoch, the signal from every channel was processed individually. The PSD was computed using the Welch algorithm. We then defined five canonical frequency bands—delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), spindle (12 - 14 Hz) and beta (14–30 Hz). For each band, the mean PSD within the corresponding frequency range was calculated by averaging the PSD values across that band. This process was repeated across all channels and frequency bands, resulting in a feature vector of size C $\times$ B per epoch, where C is the number of EEG channels and B is the number of frequency bands.

C. Epilepsy Datasets

The CHB-MIT Scalp EEG Database comprises EEG recordings from 22 pediatric subjects (5 males, aged 3–22 years, and 17 females, aged 1.5–19 years) with intractable seizures. Subjects were monitored over multiple days following the withdrawal of anti-seizure medication to capture seizure activity and evaluate their suitability for surgical intervention. In total, 182 seizures were annotated with precise onset and offset timings.

All signals were sampled at 256 Hz with 16-bit resolution. Most recordings contain 23 EEG channels, though a few have 24 or 26. Electrode placement follows the International 10–20 system for EEG acquisition.

We selected the first ten subjects for our experiments. The raw EEG signals were segmented using non-overlapping sliding windows of 3 seconds. We extracted all seizure segments and randomly selected an equal number of non-seizure segments from the remaining data to ensure class balance.

For feature extraction, we employed a commonly used and effective time-domain features for seizure detection: band power. Specifically, the raw EEG signals were filtered into five standard frequency bands: δ (0.5–4 Hz), θ (4–8 Hz), α (8–13 Hz), β (13–30 Hz), and γ (30–50 Hz).

For each frequency band, the signal from every channel was individually filtered and the power was calculated as the mean squared amplitude over time. The power values from all channels were then averaged to obtain a single scalar representing the overall energy in that frequency band. This process was repeated for all five bands, resulting in a five-dimensional feature vector per EEG epoch. These band power features provide a compact summary of the frequency-specific signal energy across the scalp.

V. EXPERIMENTS

Extensive experiments are conducted in this section to validate the effectiveness of PURE. We use the Qwen-2.5-14B-Instruct-1M model as the LLM, and apply two demonstrations per category, resulting in a total of four ICL demonstrations for the MI binary classification dataset, ten for the sleep five-class classification task, and four for the epilepsy binary classification dataset.

A. Comparing Method

1) *BL*, which randomly selects all M samples.
2) *CTD*, which applies the k-means algorithm within each class and selects centroids.
3) *RD* [34], which starts by selecting the centroid of each class as the initial point. Then, iteratively performs additional rounds of clustering on each class and selects the centroid of the largest cluster not yet covered by the previously selected points.
4) *SVM-near*, which trains a SVM classifier on the training set and selects, from each class, the sample closest to the decision boundary.
5) *SVM-far*, which in contrast to the above method, selects the sample farthest to the decision boundary within each class. To mitigate the impact of outliers, the method further filters candidates by checking the proportion of same-label samples among their five nearest neighbors.
6) *PURE*, which selects the highest-purity samples from each class as describe in section III-B

B. Experimental Setup

To comprehensively evaluate our method, we conducted experiments under three different settings. For the MI datasets, we adopted (i) within-subject, cross-session evaluation and (ii) cross-subject evaluation. For the sleep and epilepsy dataset, as only one session per subject is available, we employed (iii) within-subject five-fold cross-validation instead of (i). For the epilepsy dataset, due to the large variation in the number of seizure segments across subjects, cross-subject experimental results may be affected. Therefore, we only conduct within-subject experiments.

In the first two settings, each method was repeated five times to ensure stability. Due to the high variability introduced by random selection, experiments involving random baselines were repeated twenty times. For five-fold cross-validation experiments, the entire five-fold process was repeated five times, resulting in a total of 25 runs per method. The average classification accuracy was used as the performance metric.

1) *Within-Subject Cross-Session:* In this setting, we used the training and testing splits predefined by each dataset. This approach avoids session leakage while ensuring realistic session-wise generalization. However, the ratio between training and testing samples varies across datasets. Specifically:
   - BNCI2014001: One session is used for training and another for testing, each containing 144 trials.
   - BNCI2014004: Five sessions are available, with the first three used for training and the last two for testing.
   - BNCI2014002: A single session with eight runs; the first five runs are used for training and the remaining three for testing. Each run contains 10 trials for right-hand (RH) and 10 for feet imagery.
   - BNCI2015001: Most subjects have one training session and one testing session, totaling 400 trials. Subjects 8 and 9 have three sessions, where the first two are used for training and the last one for testing.
   - Weibo2014: This dataset contains only one session, but since the distribution and quantity of both classes are balanced between the first and second halves, the first half is used for training and the second half for testing.
2) *Within-Subject five-Fold Cross-Validation:* For this setting, we employed the `StratifiedKFold` utility from Scikit-learn. All trials of each subject were shuffled and split into five folds, ensuring that the class proportions in each fold matched those of the full dataset. This approach also maintains a consistent ratio of training to testing samples across all datasets.
3) *Cross-Subject:* A leave-one-subject-out five-fold cross-validation strategy was adopted. In each fold, one subject was reserved for testing, and the remaining subjects were combined as the training set. To mitigate inter-subject variability in EEG distributions, Euclidean Alignment (EA) [5] was applied to the data from each subject.

C. Results Analysis

Tables II, III, IV, V, VI, VII and VIII presents the classification accuracies on the seven EEG datasets, respectively. We can observe that:

1) Active selection of ICL demonstrations significantly outperforms random selection. In both within-subject and cross-subject settings, actively selected demonstrations consistently lead to a better prompting effect on the LLM, resulting in higher prediction accuracy and more stable outputs compared to random selection.
2) High-informativeness samples yield the worst prompting effect in ICL. Contrary to conventional expectations in traditional machine learning, selecting the most informative samples leads to an inferior prompting effect on the LLM, even worse than random selection. This suggests a fundamental difference between how LLMs and traditional classifiers benefit from informative instances.
3) Low-informativeness samples achieve the best prompting effect. Demonstrations with low informativeness and high model confidence, as exemplified by SVM-far and PURE, result in superior prompting effects. We hypothesize that when classifying complex numerical data, LLMs primarily rely on statistical distributional patterns rather than constructing deep decision boundaries. Therefore, more prototypical and easily distinguishable examples are more effective in guiding the model, whereas highly ambiguous samples fail to provide meaningful distinctions.
4) PURE consistently outperforms SVM-far. Although SVM-far has been optimized to avoid outlier selection and shows competitive performance, PURE still achieves better overall prompting effects across datasets and experimental conditions.
5) PURE achieves consistent superiority across multiple datasets and paradigms. Our proposed method PURE consistently outperforms random selection across all experiments and achieves optimal results in the majority of cases. Specifically, in the within-subject setting, PURE improves over random selection by 8.04%, 8.19%, 6.03%, 3.43%, and 4.09% across the five MI datasets, by 5.44% on the sleep dataset, and by 3.68% on the epilepsy dataset. In the cross-subject setting, PURE achieves similar improvements: 10.17%, 6.32%, 5.35%, 1.37%, and 1.93% across the five MI datasets, 5.67% on the sleep dataset.
6) Representativeness- and diversity-based methods (RD and CTD) also outperform random selection Although RD and CTD perform slightly worse than PURE overall, they consistently outperform random selection across all experiments. Notably, in cross-subject settings, their performance significantly improves, achieving results comparable to or even exceeding PURE on several MI datasets. Moreover, since these methods do not strictly rely on label information (labels are only used to maintain class balance), they exhibit broader applicability to scenarios where label information is limited or unavailable.

D. Impact of the Number of Demonstration Samples

In the preceding experiments, we selected two samples per category as demonstrations. In this subsection, we in-

TABLE II
WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON BNCI2014001. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
Within-subject classification										
BL	49.31	49.03	67.64	56.11	48.19	55.28	52.22	56.81	83.47	57.56
RD	55.42	48.89	77.92	64.44	48.06	53.19	54.31	71.11	80.14	61.50
CTD	56.81	44.17	71.53	62.92	50.28	51.81	54.58	76.25	86.11	61.61
SVM-near	38.89	50.00	45.00	53.06	50.56	47.36	48.19	46.11	71.53	50.08
SVM-far	58.89	47.64	75.00	61.11	52.78	55.69	57.22	73.06	85.00	<u>62.93</u>
PURE	67.08	51.94	68.61	68.75	52.50	59.44	58.19	77.22	86.67	**65.60 (↑8.04)**
Cross-subject classification										
BL	48.68	50.42	66.25	59.93	53.33	55.83	45.69	53.68	38.54	52.48
RD	61.25	51.18	74.44	64.65	57.50	61.81	67.64	70.69	52.85	<u>62.45</u>
CTD	58.13	50.14	63.54	63.82	56.53	54.03	63.47	74.44	76.53	62.29
SVM-near	37.15	47.50	52.99	59.65	48.82	43.26	36.11	42.43	26.67	43.84
SVM-far	60.76	47.85	66.67	61.18	54.37	61.53	58.75	72.08	74.10	61.92
PURE	60.76	52.15	70.07	60.76	54.79	62.92	62.64	68.89	70.83	**62.65 (↑10.17)**

TABLE III
WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON BNCI2014002. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	Avg.
Within-subject classification															
BL	61.33	75.00	72.67	60.67	59.33	68.33	84.00	64.00	88.00	54.67	47.67	57.00	47.33	52.67	63.76
RD	65.00	74.67	76.67	68.00	60.67	84.00	89.67	71.67	88.67	56.33	73.67	47.33	49.00	57.67	68.79
CTD	66.67	78.67	80.33	61.33	66.00	83.00	90.00	51.67	94.33	56.00	67.00	54.33	45.33	54.00	67.76
SVM-near	37.33	47.67	20.67	47.67	41.00	89.00	59.67	60.00	72.33	53.33	81.33	61.67	44.33	56.67	55.19
SVM-far	71.00	69.33	77.33	75.33	64.67	82.33	86.67	70.00	92.33	54.00	81.00	61.33	50.33	52.33	<u>70.57</u>
PURE	63.33	74.67	88.33	77.67	61.67	83.00	88.67	71.67	90.00	55.00	79.00	70.00	46.33	58.00	**71.95 (↑8.19)**
Cross-subject classification															
BL	55.50	60.50	60.62	56.75	55.88	62.38	60.63	62.63	80.25	61.88	59.50	55.75	56.13	48.75	59.80
RD	56.63	69.00	86.75	61.37	67.50	71.88	85.50	70.37	61.25	64.50	61.75	65.00	57.25	44.75	65.96
CTD	55.63	68.75	85.88	63.00	66.63	73.00	85.25	67.00	82.25	65.25	62.25	67.62	55.50	48.88	**67.64 (↑7.84)**
SVM-near	56.50	29.38	26.75	45.88	40.00	67.37	32.00	66.87	36.75	45.50	63.87	38.12	44.12	53.25	46.17
SVM-far	56.50	64.50	81.75	57.13	59.12	67.87	81.75	61.50	83.63	56.37	62.50	60.88	53.75	53.38	64.33
PURE	57.38	66.25	86.75	55.13	62.75	69.12	85.00	62.38	85.13	60.75	65.50	66.25	51.38	51.88	<u>66.12</u>

TABLE IV
WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON BNCI2014004. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
Within-subject classification										
BL	51.44	55.00	51.00	65.31	54.94	47.62	50.44	58.94	51.31	54.00
RD	55.44	49.57	55.40	75.25	54.00	62.62	52.81	71.50	52.81	58.82
CTD	49.19	56.93	52.87	65.81	50.00	52.19	54.88	56.94	54.31	54.79
SVM-near	49.31	56.93	51.40	47.25	51.25	45.94	42.87	52.75	42.81	48.95
SVM-far	53.25	56.29	54.00	77.25	55.50	54.69	49.75	75.94	57.06	<u>59.30</u>
PURE	51.67	50.14	53.67	76.88	61.44	57.25	52.87	78.25	58.13	**60.03 (↑6.03)**
Cross-subject classification										
BL	50.92	49.79	50.56	54.73	51.84	49.83	51.53	52.13	54.44	51.75
RD	54.89	52.35	50.44	64.81	50.19	57.50	50.14	56.76	48.17	53.92
CTD	47.00	48.26	50.11	45.08	46.73	51.61	49.72	46.76	52.56	48.65
SVM-near	45.36	48.85	49.14	41.84	47.35	48.33	48.17	59.89	48.67	48.62
SVM-far	59.11	52.88	50.00	68.27	53.14	55.19	52.00	61.47	52.36	<u>56.05</u>
PURE	57.00	52.56	52.64	69.05	55.51	57.14	53.25	64.76	52.03	**57.10 (↑5.35)**

TABLE V

WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON BNCI2015001. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Within-subject classification										
Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
BL	74.30	88.30	70.50	82.10	65.00	51.30	76.80	55.00	58.40	69.08
RD	79.60	89.00	55.90	77.40	70.40	54.90	76.50	58.30	54.30	68.48
CTD	79.00	89.60	55.50	78.50	69.30	56.00	74.50	58.30	50.50	67.91
SVM-far	67.00	86.30	83.00	82.60	67.70	52.20	81.70	64.90	62.40	71.98
SVM-near	66.70	60.80	53.00	28.60	40.80	50.40	41.20	45.80	51.60	48.77
PURE	71.80	88.80	70.80	82.80	74.40	51.20	76.00	69.20	67.60	**72.51 (↑3.43)**
Cross-subject classification										
Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
BL	70.25	80.15	68.45	65.55	61.85	56.40	55.35	56.60	64.03	64.29
RD	78.50	83.55	65.85	55.60	68.75	56.65	62.00	56.63	63.27	65.64
CTD	71.90	79.35	66.85	62.70	68.35	55.75	58.50	57.87	63.77	65.00
SVM-far	76.15	76.80	66.00	66.20	62.70	50.15	56.70	56.67	58.67	63.34
SVM-near	73.95	69.45	55.25	54.40	69.00	57.45	57.10	56.77	60.03	61.49
PURE	76.70	85.30	66.15	66.45	63.50	56.25	60.50	53.70	62.40	**65.66 (↑1.37)**

TABLE VI

WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON WEIBO2014. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Within-subject classification											
Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Average
BL	67.50	47.50	70.25	53.00	66.50	68.29	56.25	88.50	84.50	64.50	66.68
RD	67.75	46.50	73.00	59.25	76.50	68.57	63.75	84.00	84.75	71.75	69.58
CTD	74.75	49.50	71.50	54.00	69.75	69.71	55.75	84.00	83.50	67.25	67.97
SVM-far	76.25	59.00	73.50	48.50	67.50	75.43	61.50	77.75	90.00	69.25	69.87
SVM-near	37.75	45.75	60.25	46.25	44.25	49.14	43.50	43.75	88.75	63.00	52.24
PURE	69.50	61.75	76.00	67.50	69.25	71.71	57.25	80.00	87.50	67.25	**70.77 (↑4.09)**
Cross-subject classification											
Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Average
BL	63.69	54.06	62.32	52.19	62.94	63.07	73.07	78.25	78.25	61.50	64.93
RD	71.13	52.37	66.87	51.38	64.50	64.14	75.75	80.37	87.75	64.87	**67.91 (↑2.98)**
CTD	72.94	48.75	66.75	51.88	62.25	55.86	77.62	63.75	85.75	66.25	65.18
SVM-far	68.37	54.13	72.12	57.87	64.12	63.29	71.12	80.50	83.37	60.38	67.53
SVM-near	63.00	53.75	35.88	46.88	54.87	68.71	58.25	43.00	26.37	60.38	51.11
PURE	69.75	55.13	71.37	55.13	60.50	60.00	73.50	74.25	89.38	59.50	66.86

TABLE VII

WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON SLEEP-EDF. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Within-subject classification								
Methods	S1	S2	S3	S4	S5	S6	S7	Average
BL	29.81	30.93	31.93	28.26	35.18	30.18	31.87	31.17
RD	26.31	28.76	36.58	29.13	34.81	32.76	35.06	31.92
CTD	23.92	27.08	34.76	26.28	35.66	29.38	28.30	29.34
SVM-far	37.83	28.43	36.49	31.51	39.63	39.25	37.95	35.87
SVM-near	28.65	25.85	27.47	26.07	31.34	31.32	26.40	28.16
PURE	35.57	33.93	35.71	35.97	50.78	31.48	32.80	**36.61 (↑5.44)**
Cross-subject classification								
BL	23.59	21.81	19.52	24.87	27.20	20.36	23.18	22.93
RD	22.67	24.41	17.25	18.27	22.47	18.01	22.15	20.89
CTD	21.49	20.47	19.62	18.79	19.52	18.10	25.96	20.85
SVM-far	24.81	23.24	17.98	19.89	22.68	16.16	32.36	22.73
SVM-near	20.44	21.56	16.13	27.30	22.83	30.13	23.01	23.06
PURE	29.79	25.81	26.28	20.37	34.64	34.66	24.63	**28.60 (↑5.67)**

TABLE VIII
WITHIN-SUBJECT CLASSIFICATION ACCURACIES (%) ON CHB-MIT. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
BL	84.93	84.35	81.25	59.68	84.52	51.85	82.82	74.88	90.43	77.12
CTD	71.00	73.22	79.27	41.59	87.44	48.67	85.18	79.29	90.71	71.45
RD	55.40	70.44	78.41	44.27	87.02	51.07	83.64	74.27	88.71	70.43
SVM-near	39.87	68.53	73.33	59.88	79.08	42.17	79.91	67.99	89.57	66.87
SVM-far	81.33	83.70	79.15	74.16	81.10	58.37	80.00	74.39	89.57	<u>77.35</u>
PURE	85.40	86.03	82.98	72.64	86.17	64.20	83.55	76.27	89.43	**80.80 (↑3.68)**

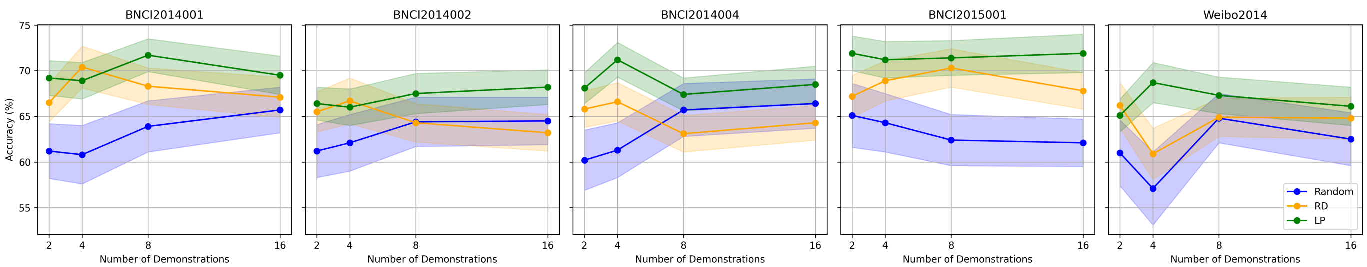

Fig. 3. Performance under different numbers of in-context demonstrations on five MI datasets using BL, RD, and PURE methods. The LLM used here is our base model: Qwen-2.5-14b-instruct-1m.

vestigate the impact of varying the number of demonstration samples on performance. The experiments are conducted on the BNCI2014001 dataset. Figure 3 presents the performance trends as the number of demonstration samples increases, from which we observe the following:

1) The performance of the LLM improves as the number of demonstration samples increases.
2) As the number of demonstration samples increases, the marginal performance gains progressively decrease, and the results across different selection methods tend to converge.
3) The results of random selection exhibit large fluctuations across repeated trials, indicating significant instability. This further reflects the sensitivity of LLMs to the choice of demonstrations.
4) PURE consistently outperforms random and other selection methods across different numbers of demonstrations, and exhibits the smallest variance across repeated experiments, indicating the highest stability.

E. Impact of Different Models

To further assess the generalizability of the proposed method, we evaluated its performance on alternative LLMs. Specifically, DeepSeek-V3 was selected for comparative analysis. Experiments were conducted on the BNCI2015-002 dataset, and the corresponding results are presented in Table IX. The following observations can be made:

1) Active selection methods consistently outperform random selection across different LLMs.
2) The relative performance hierarchy among all methods remains consistent with previous experiments, confirming the robustness of active selection strategies across different model architectures.
3) Employing larger and more powerful models leads to a consistent improvement in accuracy for all methods.Remarkably, on certain datasets, the results surpass the SOTA performance achieved by conventional deep neural networks, highlighting the considerable potential of LLM-based ICL few-shot learning approaches in non-text domains such as EEG signal analysis.

VI. CONCLUSIONS

This paper proposed a purity-guided unique-round example (PURE) approach for EEG signal classification using LLMs in a few-shot in-context learning scenario. Existing demonstration selection methods commonly used in NLP and CV domains are often ineffective for EEG data due to its complexity and inherent lack of semantic interpretability. PURE addresses these limitations by integrating domain-specific feature extraction with an active selection strategy based on local same-label purity, selecting compact sets of representative, low-entropy examples. Extensive experiments across multiple EEG datasets and BCI paradigms demonstrate that PURE significantly outperforms random selection and other baseline methods, achieving higher accuracy and stability. Furthermore, PURE offers substantial computational efficiency through a one-time selection strategy, highlighting its practical utility in real-world EEG classification applications.

REFERENCES

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, Long Beach, LA, Dec. 2017, pp. 5998–6008.

[2] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell *et al.*, "Language models are few-shot learners," *Advances in Neural Information Processing Systems*, vol. 33, pp. 1877–1901, Dec. 2020.

TABLE IX
WITHIN-SUBJECT CLASSIFICATION ACCURACIES (%) ON BNCI2015001 USING THE DEEPSEEK-V3 MODEL. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Methods	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
BL	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	51.00
CTD	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	51.00
RD	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	51.00
SVM-near	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	51.00
SVM-far	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	51.00
PURE	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	51.00

[3] B. J. Lance, S. E. Kerick, A. J. Ries, K. S. Oie, and K. McDowell, "Brain-computer interface technologies in the coming decades," *Proc. of the IEEE*, vol. 100, no. 3, pp. 1585–1599, 2012.

[4] J. R. Wolpaw, N. Birbaumer, D. J. McFarland, G. Pfurtscheller, and T. M. Vaughan, "Brain-computer interfaces for communication and control," *Clinical Neurophysiology*, vol. 113, no. 6, pp. 767–791, 2002.

[5] H. He and D. Wu, "Transfer learning for brain–computer interfaces: A euclidean space data alignment approach," *IEEE Trans. on Biomedical Engineering*, vol. 67, no. 2, pp. 399–410, 2019.

[6] S. Hegselmann, A. Buendia, H. Lang, M. Agrawal, X. Jiang, and D. Sontag, "Tabllm: Few-shot classification of tabular data with large language models," in *Int'l Conf. on Artificial Intelligence and Statistics*, Valencia, Spain, Apr. 2023, pp. 5549–5581.

[7] X. Liu, D. McDuff, G. Kovacs, I. Galatzer-Levy, J. Sunshine, J. Zhan, M.-Z. Poh, S. Liao, P. Di Achille, and S. Patel, "Large language models are few-shot health learners," *arXiv preprint arXiv:2305.15525*, 2023.

[8] J. Decety, "The neurophysiological basis of motor imagery," *Behavioural brain research*, vol. 77, no. 1-2, pp. 45–52, 1996.

[9] K. A. I. Aboalayon, M. Faezipour, W. S. Almuhammadi, and S. Moslehpour, "Sleep stage classification using eeg signal analysis: a comprehensive survey and new investigation," *Entropy*, vol. 18, no. 9, p. 272, 2016.

[10] U. R. Acharya, S. V. Sree, G. Swapna, R. J. Martis, and J. S. Suri, "Automated eeg analysis of epilepsy: a review," *Knowledge-Based Systems*, vol. 45, pp. 147–165, 2013.

[11] H. Ramoser, J. Muller-Gerking, and G. Pfurtscheller, "Optimal spatial filtering of single trial EEG during imagined hand movement," *IEEE Trans. on Rehabilitation Engineering*, vol. 8, no. 4, pp. 441–446, 2000.

[12] A. R. Hassan and M. I. H. Bhuiyan, "Automated identification of sleep states from eeg signals by means of ensemble empirical mode decomposition and random under sampling boosting," *Computer methods and programs in biomedicine*, vol. 140, pp. 201–210, 2017.

[13] D. R. Cox, "The regression analysis of binary sequences," *Journal of the Royal Statistical Society Series B: Statistical Methodology*, vol. 20, no. 2, pp. 215–232, 1958.

[14] C. Cortes and V. Vapnik, "Support-vector networks," *Machine learning*, vol. 20, pp. 273–297, 1995.

[15] K. K. Ang, Z. Y. Chin, H. Zhang, and C. Guan, "Filter bank common spatial pattern (FBCSP) in brain–computer interface," in *IEEE Int'l Joint Conference on Neural Networks*, Hong Kong, China, Jun. 2008, pp. 2390–2397.

[16] F. Lotte and C. Guan, "Regularizing common spatial patterns to improve bci designs: unified theory and new algorithms," *IEEE Transactions on biomedical Engineering*, vol. 58, no. 2, pp. 355–362, 2010.

[17] A. Craik, Y. He, and J. L. Contreras-Vidal, "Deep learning for electroencephalogram (eeg) classification tasks: a review," *Journal of neural engineering*, vol. 16, no. 3, p. 031001, 2019.

[18] A. Shoeibi, M. Khodatars, N. Ghassemi, M. Jafari, P. Moridian, R. Alizadehsani, M. Panahiazar, F. Khozeimeh, A. Zare, H. Hosseini-Nejad *et al.*, "Epileptic seizures detection using deep learning techniques: a review," *Int'l journal of environmental research and public health*, vol. 18, no. 11, p. 5780, 2021.

[19] R. T. Schirrmeister, J. T. Springenberg, L. D. J. Fiederer, M. Glasstetter, K. Eggensperger, M. Tangermann, F. Hutter, W. Burgard, and T. Ball, "Deep learning with convolutional neural networks for eeg decoding and visualization," *Human brain mapping*, vol. 38, no. 11, pp. 5391–5420, 2017.

[20] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: a compact convolutional neural network for EEG-based brain-computer interfaces," *Journal of Neural Engineering*, vol. 15, no. 5, p. 056013, 2018.

[21] X. Jiang, L. Meng, X. Chen, Y. Xu, and D. Wu, "Csp-net: Common spatial pattern empowered neural networks for eeg-based motor imagery classification," *Knowledge-Based Systems*, vol. 305, p. 112668, 2024.

[22] A. Supratak, H. Dong, C. Wu, and Y. Guo, "Deepsleepnet: A model for automatic sleep stage scoring based on raw single-channel eeg," *IEEE transactions on neural systems and rehabilitation engineering*, vol. 25, no. 11, pp. 1998–2008, 2017.

[23] M. Golmohammadi, S. Ziyabari, V. Shah, E. Von Weltin, C. Campbell, I. Obeid, and J. Picone, "Gated recurrent networks for seizure detection," in *Proceedings of the IEEE Signal Processing in Medicine and Biology Symposium (SPMB)*, Philadelphia, PA, USA, Dec. 2017, pp. 1–6.

[24] W. Jiang, L. Zhao, and B. liang Lu, "Large brain model for learning generic representations with tremendous EEG data in BCI," in *The Twelfth Int'l Conf. on Learning Representations*, Vienna, Austria, May. 2024.

[25] B. Settles, "Active learning literature survey," University of Wisconsin–Madison, Computer Sciences Technical Report 1648, 2009.

[26] D. D. Lewis, "A sequential algorithm for training text classifiers: Corrigendum and additional data," in *Acm Sigir Forum*, vol. 29, no. 2. ACM New York, NY, USA, 1995, pp. 13–19.

[27] H. T. Nguyen and A. Smeulders, "Active learning using pre-clustering," in *Proc. of the twenty-first int'l conf. on Machine learning (ICML 2004)*, Banff, Alberta, Canada, Jul. 2004, p. 79.

[28] B. Settles and M. Craven, "An analysis of active learning strategies for sequence labeling tasks," in *Proc. of the 2008 conf. on empirical methods in natural language processing*, Honolulu, Hawaii, October 2008, pp. 1070–1079.

[29] D. Shen, J. Zhang, J. Su, G. Zhou, and C.-L. Tan, "Multi-criteria-based active learning for named entity recognition," in *Proc. 42nd Annual Meeting on Association for Computational Linguistics*, Barcelona, Spain, July 2004.

[30] S.-J. Huang, R. Jin, and Z.-H. Zhou, "Active learning by querying informative and representative examples," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 36, no. 10, pp. 1936–1949, 2014.

[31] W. Cai, Y. Zhang, S. Zhou, W. Wang, C. Ding, and X. Gu, "Active learning for support vector machines with maximum model change," *Lecture Notes in Computer Science*, vol. 8724, pp. 211–216, 2014.

[32] T. He, S. Zhang, J. Xin, P. Zhao, J. Wu, X. Xian, C. Li, and Z. Cui, "An active learning approach with uncertainty, representativeness, and diversity," *The Scientific World Journal*, 2014.

[33] R. Burbidge, J. J. Rowland, and R. D. King, "Active learning for regression based on query by committee," in *Proc. IEEE Int'l Conf. on Computer Vision*, vol. 4881, Venice, Italy, Oct. 2017, pp. 5286–5295.

[34] D. Wu, "Pool-based sequential active learning for regression," *IEEE Trans. on Neural Networks and Learning Systems*, vol. 30, no. 5, pp. 1348–1359, 2018.

[35] D. Wu, C.-T. Lin, and J. Huang, "Active learning for regression using greedy sampling," *Information Sciences*, vol. 474, pp. 90–105, 2019.

[36] A. Chowdhery, S. Narang, J. Devlin, M. Bosma, G. Mishra, A. Roberts, P. Barham, H. W. Chung, C. Sutton, S. Gehrmann *et al.*, "Palm: Scaling language modeling with pathways," *Journal of Machine Learning Research*, vol. 24, no. 240, pp. 1–113, 2023.

[37] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat *et al.*, "Gpt-4 technical report," *arXiv preprint arXiv:2303.08774*, 2023.

[38] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar *et al.*, "Llama: Open and efficient foundation language models," *arXiv preprint arXiv:2302.13971*, 2023.

[39] Q. Dong, L. Li, D. Dai, C. Zheng, J. Ma, R. Li, H. Xia, J. Xu, Z. Wu, B. Chang, X. Sun, L. Li, and Z. Sui, "A survey on in-context learning,"

in *Proc. of the 2024 Conf. on Empirical Methods in Natural Language Processing*, Miami, FL, Nov. 2024, pp. 1107–1128.

[40] M. Luo, X. Xu, Y. Liu, P. Pasupat, and M. Kazemi, "In-context learning with retrieved demonstrations for language models: A survey," *arXiv preprint arXiv:2401.11624*, 2024.

[41] J. Liu, D. Shen, Y. Zhang, B. Dolan, L. Carin, and W. Chen, "What makes good in-context examples for GPT-3?" in *Proc. of Deep Learning Inside Out (DeeLIO 2022): The 3rd Workshop on Knowledge Extraction and Integration for Deep Learning Architectures*, Dublin, Ireland, May. 2022, pp. 100–114.

[42] J. Li, J. Wang, Z. Zhang, and H. Zhao, "Self-prompting large language models for zero-shot open-domain QA," in *Proc. of the 2024 Conf. of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Mexico City, Mexico, Jun. 2024, pp. 296–310.

[43] A. Scarlatos and A. Lan, "Reticl: Sequential retrieval of in-context examples with reinforcement learning," *arXiv preprint arXiv:2305.14502*, 2023.

[44] Y. Zhang, K. Zhou, and Z. Liu, "What makes good examples for visual in-context learning?" vol. 36, New Orleans, LA, Dec. 2023, pp. 17 773–17 794.

[45] S. Hongjin, J. Kasai, C. H. Wu, W. Shi, T. Wang, J. Xin, R. Zhang, M. Ostendorf, L. Zettlemoyer, N. A. Smith *et al.*, "Selective annotation makes language models better few-shot learners," in *The Eleventh Int'l Conf. on Learning Representations*, Kigali, Rwanda, May. 2023.

[46] P. Boonyakitanont, A. Lek-Uthai, K. Chomtho, and J. Songsiri, "A review of feature extraction and performance evaluation in epileptic seizure detection using eeg," *Biomedical Signal Processing and Control*, vol. 57, p. 101702, 2020.

[47] B. Chen, Z. Zhang, N. Langrené, and S. Zhu, "Unleashing the potential of prompt engineering in large language models: a comprehensive review," *arXiv preprint arXiv:2310.14735*, 2023.

[48] V. Jayaram and A. Barachant, "Moabb: trustworthy algorithm benchmarking for bcis," *Journal of Neural Engineering*, vol. 15, no. 6, p. 066011, 2018.

[49] M. Tangermann, K.-R. Müller, A. Aertsen, N. Birbaumer, C. Braun, C. Brunner, R. Leeb, C. Mehring, K. J. Miller, G. R. Müller-Putz *et al.*, "Review of the BCI competition IV," *Frontiers in Neuroscience*, vol. 6, p. 55, 2012.

[50] J. Faller, C. Vidaurre, T. Solis-Escalante, C. Neuper, and R. Scherer, "Autocalibration and recurrent adaptation: Towards a plug and play online erd-bci," *IEEE Trans. on Neural Systems and Rehabilitation Engineering*, vol. 20, no. 3, pp. 313–319, 2012.